Richard Charette

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Professional Summary

Skilled mathematician/scientist with expertise in topology, abstract algebra, differential geometry, and computational physics. Proficient in MATLAB, Python, and C/C++ with extensive experience in physics-based software, machine learning, and biological data analysis.

Clearance

Active Secret Clearance

Professional Experience

Raytheon, Marlborough, MA Jul 2025 – Present
Systems Engineer – Radar Systems

- Support integration and verification of radar subsystems, ensuring requirements traceability and compliance across hardware and software components.
- Analyze radar system performance using MATLAB and Python, modeling signal processing chains and validating system behavior under real-world conditions.

BAE Systems/Donatech, Hudson, NH Nov 2024 – Feb 2025
Image Processing Software Engineer/Mathematician

- Designed and implemented Cooley-Tukey FFT algorithms for efficient decomposition of large FFTs, utilizing C++ object-oriented programming to structure complex software components in a Linux environment.
- Developed OpenCL-based high-throughput compute kernels for FPGA acceleration in radar signal processing, incorporating multi-threading and synchronization for parallel execution.
- Integrated algorithms into existing systems and debugged with Linux tools to ensure functionality.

NAVSEA, Newport, RI Sep 2022 – Feb 2023
Software Engineer/Staff Scientist

- Provided technical expertise and developed C++ software components for submarine communication system development, focusing on real-time integration and deployment in a Linux environment.
- Collaborated with engineers and scientists to define system requirements, using MATLAB to model antenna data and Python for additional analysis, adapting to new tools as needed.

Lockheed Martin, Marion, MA May 2021 – Feb 2022
Software Engineer/Manufacturing Engineer

- Developed CAD schematics for SUBCOMs and wiring systems, integrating C++-based machine vision software in a Linux environment to identify cable shortages, which drove a site-wide redesign.
- Contributed to a research team, using GitHub for version control and RabbitMQ/StormMQ for distributed coordination, debugging a machine learning model to pinpoint algorithmic bias.
- Adapted to SolidWorks and 3D printing workflows, drafting new models, and collaborating with manufacturing teams to implement updated designs.

University of Massachusetts Dartmouth, Dartmouth, MA Aug 2019 – May 2021
Graduate Researcher

- Researched Type Ia supernovae dynamics, developing C++ and Python simulation wrappers in a Linux environment to model star sequences and the Chandrasekhar limit, utilizing multi-threading for data processing efficiency.

Fellowships

Tufts University Mar 2023 – Jun 2024
Fellowship

- Applied machine learning models to analyze biological data using a database with Astropath, developing C++11 and Python scripts in a Linux environment to process raw data and perform differential gene expression analysis, incorporating parallel programming for efficiency.
- Prepared gene-level data for clustering and visualization (UMAP, t-SNE, heatmaps), adapting to new bioinformatics tools and collaborating with researchers via GitHub.
- Analyzed a GeoMx mouse lungs and kidney dataset with the Whole Human Transcriptome Atlas, profiling diabetic kidney disease and healthy samples.

Wolfram Summer School Jun 2022 – Jul 2022
Fellowship

- Enumerated axiom systems and computed entailment cones using Wolfram Mathematica, quickly adapting to its symbolic computation framework to model topological properties.
- Investigated proof paths to study topological properties using homology and homotopy theory.

Boston University/Mathematics Department, Boston, MA Aug 2017 – May 2018
Fellowship

- Analyzed neuronal spike-LFP relationships, developing C++ and MATLAB tools in a Linux environment to model stimuli responses with Fourier analysis, utilizing parallel processing to filter noise.
- Programmed a custom compiler and filter, adapting to Simulink for real-time simulation and collaborating with neuroscientists to validate models.

Biogen Idec Inc., Cambridge, MA May 2010 – Aug 2010
Fellowship

- Investigated gene transfer in E. coli for bioluminescence, utilizing MATLAB to analyze data and produce summaries while adapting to gene editing tools and team workflows.
- Collaborated with researchers to refine experimental methods, presenting results to supervisors for biotechnological insights.

Technical Skills

Programming Languages: MATLAB, Python, C/C++, C++11, FORTRAN, R, APL, Wolfram
Software & Tools: GitHub, Linux/Unix, LaTeX, Mathematica, Wolfram Mathematica, STATA, R Studio, RabbitMQ, StormMQ, AutoCAD, SolidWorks, 3D Printing Modeling, Moby, TORCH, SciPy, NumPy, Simulink, Git, Software Documentation, BitHub, Astropath, Slack
Technical Domains: Image Processing, Machine/Computer Vision, Artificial Neural Networks, FPGA Acceleration, Radar Signal Processing, Submarine Communication Systems, Antenna Systems, Object-Oriented Programming, Integrated Systems, Large Data Computation, Structured Programming, Firmware, OpenCL, CUDA, Real-Time Software/Hardware, Signal Processing, x86 Processing, Machine Learning, Biological Data Analysis
Mathematical Concepts: Topology, Abstract Algebra, Cryptography, Differential Geometry, Number Theory, Combinatorics, Discrete Geometry, Numerical Analysis, Complex Systems, Axiom Systems, Fourier Series Equations, Fourier Transformations, Proof Paths, Proof Space, Two-Way Proofs
Physics Concepts: Statistical Mechanics, Stellar Physics

Education

M.S. Computational Mathematics, John Hopkins University, August 2026-Present
GPA: N/A

M.S. Physics, University of Massachusetts Dartmouth, *Magna Cum Laude* May 2021
GPA: 3.79

B.S. Mathematics, University of Massachusetts Dartmouth, *Magna Cum Laude* May 2017
GPA: 3.54

B.S. Economics, Bridgewater State University, *Cum Laude* May 2013
GPA: 3.33

Grants & Awards

- Bits to Bytes Grant (2023-2024)
- Massachusetts Space Grant Consortium (2020-2021)
- Dean's List, University of Massachusetts Dartmouth (Graduate) (2019-2021)
- Dean's List, University of Massachusetts Dartmouth (Undergraduate) (2015-2017)
- Dean's List, Bridgewater State University (2011-2013)
- John and Abigail Adams Scholarship (2011-2013)
- Adrian Tinsley Undergraduate Research Scholarship (2012)
- Presenter, National Conference for Undergraduate Research (2012)